ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(b)** 2 : 1	75/25 = 3; 60/40 = 1.5; ratio 2 : 1.
3	**(d)** 4.4 g	Mass of gas escaped.
4	**(c)** A is true but R is false.	A is true; R is false because the gas escapes.
5	**(b)** 3 : 8, CO_2	Atoms ratio 2.4/12 : 6.4/16 = 1 : 2.
6	**(d)** 78.5	52.5/47.5 x 71 = 78.5.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** 28 g	28/44 of the mass is nitrogen.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Since the number and kind of atoms stay the same, the total mass is conserved.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the fixed ratio is explained by the law of constant proportions, not just by water being a compound.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	4 g H needs 32 g O (1); Water = 36 g; 8 g of oxygen left (1).	1+1
12	Na+: 2, 8; F-: 2, 8 (1); Same as neon (1).	1+1
13	117/58.5 = 2 (1); Twice as many (1).	1+1
14	Formula unit mass of CuO = 64 + 16 = 80 u (1); So 64 g of copper is present in 80 g (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	S: 1/32 = 0.03125 (1); O: 1.6/16 = 0.1 (1); Ratio about 1 : 3: SO_3 (1).	1+1+1
16	Mg2+ and O2- attract more strongly (1); More energy to break the lattice (1); Hence higher melting point (about 2800 vs 801 degrees Celsius) (1).	1+1+1
17	Molecular mass = 12 + 4 = 16 u (1); Carbon = 12/16 x 100 = 75 per cent (1); Hydrogen = 4/16 x 100 = 25 per cent (1).	1+1+1
	OR: Mass of O2 = 24.5 - 14.9 = 9.6 g (1); Percentage = 9.6/24.5 x 100 (1); = 39.2 per cent (1).	1+1+1
18	Weighing error (1); Leakage of a gas (1); The law holds when all products are weighed (1).	1+1+1
	OR: Water: H 11.1, O 88.9 (1); H2O2: H 5.9, O 94.1 (1); Different compounds, different fixed compositions (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) In a chemical reaction the total mass of the reactants equals the total mass of the products (mass is neither created nor destroyed) (1). (ii) $10 - 5.6 = 4.4$ g (1). (iii) Antoine Lavoisier (1). (iv) No (1); carbon dioxide escapes into the air from the open dish, and if its mass is included the total mass stays 10 g (1).	1+1+1+2
20	Conservation: atoms rearranged (1); Constant proportions: fixed ratios of atoms (1); Atoms are divisible (1); Isotopes differ in mass (1); Atoms combine in whole-number ratios remains valid (1).	1+1+1+1+1
	OR: Mass of oxygen = 36 - 4 = 32 g (1); Total mass of products = 4 + 32 = 36 g = mass of reactant (1); Mass is conserved (Lavoisier) (1); Ratio H : O = 4 : 32 (1); = 1 : 8 (1).	1+1+1+1+1

Mistakes to avoid:

- Treating the subscript in a formula as the valency, or forgetting to reduce criss-crossed numbers to the simplest ratio (writing Ca_2O_2 for calcium oxide).
- Writing FeO_3 or Al_3O_2 by swapping the wrong numbers; remember to criss-cross *valencies* (charges) and put the cation first.
- Forgetting the bracket when a polyatomic ion occurs more than once: $Ca(OH)_2$, not $CaOH_2$.

- Mixing atomicity with number of molecules: $3\,O_2$ means three molecules, six atoms; O_3 is one molecule of ozone.
- Adding the atomic mass of an element only once when the formula has a subscript or bracket (e.g. $(NH_4)_2SO_4$: N counted twice, H eight times).
- Calling $NaCl$ a 'molecule' and giving it a 'molecular mass'; for ionic compounds use formula unit mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 310 u	120 + 62 + 128.
2	**(b)** 4 g	36 x 2/18 = 4 g.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** 175.5 : 40	3 x 58.5 : 40.
5	**(d)** 20 g	40 - 20.
6	**(d)** A is false but R is true.	A is false and R is true.
7	**(d)** 48 g	3 x 16.
8	**(b)** 16 g	Mass is conserved: 40 - 24 = 16 g.
9	**(c)** A is true but R is false.	Aluminium has valency 3 and oxygen has valency 2, so criss-crossing gives Al_2O_3; R is false.
10	**(d)** A is false but R is true.	NaCl is a crystal lattice of Na^+ and Cl^- ions, not discrete molecules, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water molecules attract and separate ions (1); Kerosene cannot interact with ions (1).	1+1
12	144 + 22 + 176 (1); = 342 u (1).	1+1
13	Mass of Na2SO4 = 142 (1); 46/142 x 100 = 32.4 g (1).	1+1
14	They do not form ions (1); No free charged particles (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Each Li loses 1 electron (1); O gains 2 electrons (1); Formula unit mass 14 + 16 = 30 u (1).	1+1+1
16	(a) Atoms are neither created nor destroyed (1); (b) Atoms of given elements combine in fixed ratios (1); Their masses are fixed (1).	1+1+1
17	Products are gases (CO2 and water vapour) that escape (1); Oxygen from air also takes part (1); A closed set-up would show equal masses (1).	1+1+1
	OR: Per unit: Al 2, S 3, O 12 (1); In 5 units: Al 10, S 15, O 60 (1); Total 85 atoms (1).	1+1+1
18	Ratio Ca : Cl = 40 : 71 (1); Ca = 20 g (1); Cl = 35.5 g (1).	1+1+1
	OR: Same nuclear charge +11 (1); Na+ is smaller (loses outer shell) (1); Na: 2, 8, 1; Na+: 2, 8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Law of constant proportions (Proust) (1). (ii) $\frac{1}{9}$ (1). (iii) $27 \times \frac{1}{9} = 3$ g (1). (iv) Oxygen $= 27 \times \frac{8}{9} = 24$ g (1); the ratio would again be 1 : 8 because a compound has fixed composition whatever its source (1).	1+1+1+2
20	H : O = 1 : 8 (1); 20 g O needs 2.5 g H (1); Water = 22.5 g (1); Excess H = 17.5 g (1); Total mass = 40 g (conservation) (1).	1+1+1+1+1
	OR: Formula unit mass of CaCO3 = 100 u (1); CaO = 56 u, so 56 g of CaO (1); CO2 = 44 u, so 44 g of CO2 (1); Total products = 56 + 44 = 100 g (1); Equal to the mass of the reactant, so mass is conserved (1).	1+1+1+1+1

Mistakes to avoid:

- Treating the subscript in a formula as the valency, or forgetting to reduce criss-crossed numbers to the simplest ratio (writing Ca_2O_2 for calcium oxide).
- Writing FeO_3 or Al_3O_2 by swapping the wrong numbers; remember to criss-cross *valencies* (charges) and put the cation first.
- Forgetting the bracket when a polyatomic ion occurs more than once: $Ca(OH)_2$, not $CaOH_2$.

- Mixing atomicity with number of molecules: $3\,O_2$ means three molecules, six atoms; O_3 is one molecule of ozone.
- Adding the atomic mass of an element only once when the formula has a subscript or bracket (e.g. $(NH_4)_2SO_4$: N counted twice, H eight times).
- Calling $NaCl$ a 'molecule' and giving it a 'molecular mass'; for ionic compounds use formula unit mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $M_2(SO_4)_3$	Criss-cross.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(c)** 6	Each Mg loses 2: 3 x 2.
4	**(c)** $M_3(PO_4)_2$	Criss-cross.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(c)** Mg^{2+}	10 electrons.
7	**(a)** an ionic compound	Ionic.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Atomicity is the number of atoms in a molecule; ozone is O_3.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$2(1) + 16 = 18$ u; both statements are true but R does not explain the value 18.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fe has valency 3 and O 2 (1); Criss-cross gives 2 Fe and 3 O (1).	1+1
12	Isotopes have the same atomic number and different mass numbers (1); Atoms of one element may have different masses (1).	1+1
13	16 (1); u (unified atomic mass unit) (1).	1+1
14	Metals have few valence electrons that are easily lost (1); Non-metals have nearly complete shells and gain electrons (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(a) 14/17 = 82.4 per cent (1); (b) 28/80 = 35 per cent (1); NH3 has more nitrogen per mass (1).	1+1+1
16	Y gains 1 (Cl-) (1); Z loses 2 (Ca2+) (1); ZY_2 ionic compound (1).	1+1+1
17	Carbonate is a polyatomic ion with charge 2- (1); Two Na+ balance it (1); Net charge zero (1).	1+1+1
	OR: Diamond: giant structure, very hard, very high m.p. (1); Iodine: separate I_2 molecules, soft, sublimes (1); Different forces (1).	1+1+1
18	Four valence electrons (1); Sharing is easier than transfer of four electrons (1); Forms CH_4 (1).	1+1+1
	OR: Mass of A = 12 u, B2 = 32 u (1); Molecular mass = 44 u (1); A : B = 12 : 32 = 3 : 8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $40 + 12 + 3(16) = 100$ u (1). (ii) N: 2, H: 8, S: 1, O: 4, total 15 atoms (1). (iii) $2(14 + 4) + 32 + 64 = 36 + 96 = 132$ u (1). (iv) $\frac{28}{132} \times 100 = 21.2\%$ (1) with mass of N = $2 \times 14 = 28$ u (1).	1+1+1+2
20	CH4: 16 u, 25 per cent (1); C2H6: 30 u, 20 per cent (1); C2H4: 28 u, 14.3 per cent (1); C2H2: 26 u, 7.7 per cent (1); Hydrogen percentage decreases with unsaturation (1).	1+1+1+1+1
	OR: Weigh a clean copper strip or powder in a crucible (1); Heat strongly in air and reweigh (1); Mass of oxygen = difference (1); Calculate Cu : O ratio 63.5 : 16 (1); Compare with 4 : 1, conclude CuO (1).	1+1+1+1+1

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